Flora Phiri

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PROFESSIONAL SUMMARY

Graduate researcher pursuing an M.S. in Information Systems Technology at George Washington University, with a data analytics background spanning healthcare, industrial, and cloud domains. Skilled in building dashboards and visualizations in Power BI and Tableau, querying and transforming structured datasets in SQL, and applying statistical methods to surface actionable insights. Experienced translating complex findings for technical and nontechnical stakeholders across remote, cross-functional environments at scale.

EDUCATION

The George Washington University, School of Business — Washington, DC
Master of Science, Information Systems Technology. — *Expected December 2026*
Relevant Coursework: Cloud Foundations, Web Development, IT Project Management, Information Systems Analysis and Design

Chinhoyi University of Technology, Graduate Business School — Chinhoyi, Zimbabwe
Master of Science, Big Data Analytics. Dissertation: A Predictive Maintenance Model for Trackless Mobile Machinery in Mining Operations: A Case Study for Mupani Mine. — *December 2024*

Midlands State University, Faculty of Business Sciences — Harare, Zimbabwe
Bachelor of Commerce (Honors), Information Systems. — *December 2022*
Relevant Coursework: Operating Systems, Data Communications and Networks, Computer Security

EXPERIENCE

Cloud Infrastructure Engineer — September 2022 – January 2025
Zimbabwe Platinum Mines — *Mhondoro-Ngezi, Zimbabwe*

- Built and maintained Power BI and Tableau dashboards to monitor performance, security, and cost across more than 100 Azure subscriptions, surfacing anomalies and inconsistencies in structured operational data and contributing to 30% annual cost optimization.
- Audited approximately 6,000 directory objects synchronized via DirSync and Azure AD Connect, maintaining 99% synchronization accuracy by identifying and resolving missing or mismatched records.
- Documented infrastructure changes, security findings, and operational reports for both technical engineering teams and nontechnical business stakeholders.
- Managed identity governance, conditional access, and multifactor authentication for more than 3,000 user accounts, reducing unauthorized access by over 90%.
- Architected Kubernetes-based container orchestration, reducing deployment time by 35% and improving release reliability.
- Designed and managed Azure Virtual Desktop for 180 remote employees, collaborating daily with a distributed team.

Cloud Infrastructure Engineer Intern (Volunteer) — June 2025 – September 2025
ABC of Cloud Computing — *Remote*

- Deployed and managed Azure virtual machines in Windows environments while working fully remotely with a distributed team.
- Supported migrations from on-premises infrastructure to cloud platforms, documenting processes for technical and nontechnical audiences.
- Monitored Azure resources and supported performance, security, and cost optimization.

PROJECTS

Healthcare & Data Analytics Projects

- Diabetes Risk Prediction with Clinical Data: Built a dense neural-network classifier using clinical measurements from the Pima Indians Diabetes dataset; standardized numerical features, validated a TensorFlow/Keras model, and evaluated results with accuracy, confusion matrices, and classification reports, achieving 78.57% test accuracy.
- Predictive Maintenance Model (Master's Dissertation): Cleaned and transformed 5,000 operational records, identifying and treating missing values, outliers, and inconsistent entries prior to modeling; compared four ML models using accuracy, precision, recall, F1, and ROC-AUC; Neural Network achieved 94.2% accuracy and 95.4% ROC-AUC; presented findings and model recommendations to mine managers and key stakeholders.
- Cloud API Security and Vulnerability Assessment (Publication): Co-authored a mixed-methods study of 312 production APIs, combining automated testing, statistical analysis (chi-square, Kruskal-Wallis, CVSS scoring), and interviews with 18 practitioners to evaluate data and system security risk. Published in ScienceXplorer, Vol. 8 (2026).
- Cloud-Based Health Information System: Designed a HIPAA-conscious cloud platform for patient-record management, focusing on access control, data availability, privacy safeguards, and secure, scalable information workflows.
- Smart DMV - AI-Assisted Document Verification Platform: Collaborated on a cloud-backed pre-appointment verification platform using OCR and AI confidence scoring to assess uploaded identity and residency documents, routing low-confidence or inconsistent

cases to human staff review.

Additional Machine Learning & Research Projects

- Natural Language Processing and Text Classification: Built an NLP workflow (text normalization, tokenization, stopword handling, stemming, lemmatization) and an embedding-based LSTM model in TensorFlow/Keras for text classification.
- Handwritten-Digit Recognition with Neural Networks: Developed a TensorFlow/Keras multilayer neural network for ten-class MNIST digit recognition, achieving 95.44% test accuracy.
- Foundational Machine Learning Portfolio: Implemented regression and classification workflows (Linear/Logistic Regression, SVM, Random Forest, Gradient Boosting) across multiple datasets, applying exploratory analysis, feature preprocessing, and cross-validation.
- Monolith-to-Microservices Migration on AWS: Containerized and migrated a monolithic Node.js application into independently deployable services using Docker, Amazon ECR, ECS, and an Application Load Balancer.

SKILLS

Data Analysis & Visualization: Microsoft Excel (advanced), Power BI, Tableau, exploratory data analysis, structured dataset review, data quality and consistency checks, Matplotlib, Seaborn

Data Engineering & Databases: ETL development, data pipelines, data integration, MySQL, PostgreSQL, Azure SQL, Databricks

Programming & Automation: Python, SQL, PowerShell, Bash, Microsoft Graph REST APIs, scripting and workflow automation

Machine Learning & AI: TensorFlow, Keras, Scikit-learn, supervised learning, neural networks, LSTM, NLP, predictive modeling

Research & Evaluation: quantitative and mixed-methods research design, documentation, feature engineering, hypothesis testing, cross-validation, accuracy, precision, recall, F1, ROC-AUC, confusion matrices

Cloud & Systems: Microsoft Azure, AWS, Kubernetes, Docker, identity governance, API security, Zero Trust, Git/GitHub

Healthcare-Adjacent & Clinical Data: clinical dataset preprocessing, HIPAA-conscious system design, patient-record access control and privacy, identity/document verification workflows

CERTIFICATIONS

- Microsoft Certified: Azure Data Fundamentals
- Microsoft Certified: Azure Fundamentals
- Microsoft Certified: Azure Administrator Associate